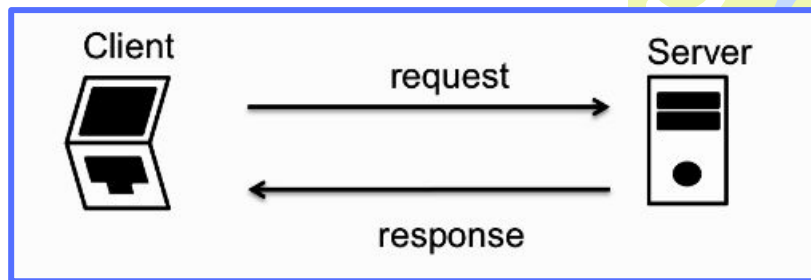


Tic-Tac-Toe Game Final Project

By: Gabby Peña, Sere Doumbouya, and
Alina Jalisi

Motivation

- Taking the classic game on paper, into a digital form
- Expanding our knowledge of how to create and use a server, as well as communicating with it
- Gain more experience on how to handle multithreading
- To gain more experience creating games in order to prepare for certain careers (such as working at Nintendo)
- Challenging ourselves
 - Managing clients, synchronization



Classes Created

01

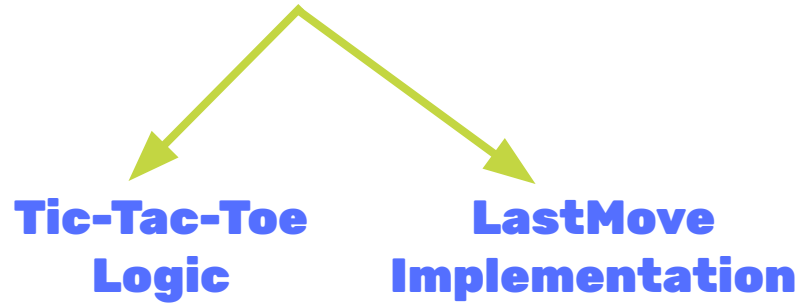
GameGUI

02

GameClient

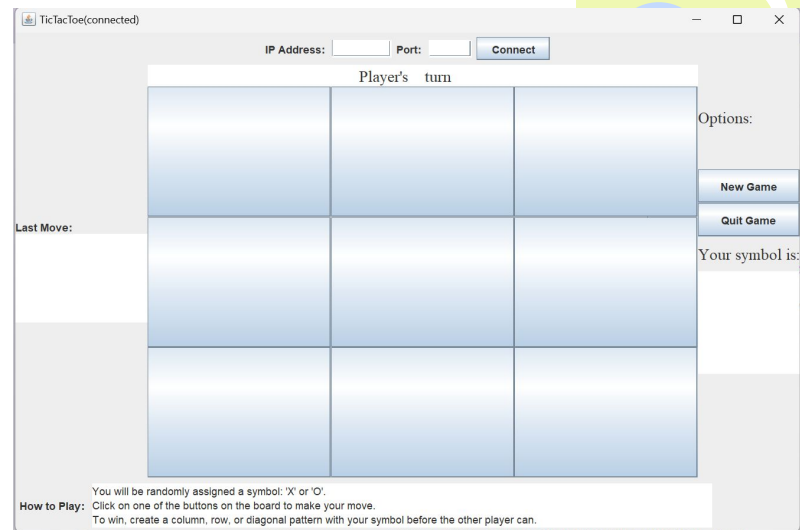
03

GameServer



GameGUI

- Code for graphical interface
 - Contains multiple panels, buttons, and text fields to make it more organized and user friendly
- Where main method is stored
- Contains error messages for when the server entered is not valid
- Allows for game board to be reset
- Controls what happens when each button is clicked
- Updates and displays what the player's symbol is (first one to join the server is X)
- When a button on the board is clicked, it updates to display that player's symbol



GameClient

- Allows a player to connect to the server given a valid IP Address and port number
- Contains a ReadingThread to read and receive moves that the players make



GameServer

- Established a socket
- Accepts connections
- Adds clients to the server
- Reads and receives messages from client
- Ensures all moves are synchronized across all GUIs



The Tic-Tac-Toe Logic

- Controls mechanisms of the actual game
- Displays a Win or Draw message depending on the outcome of the game
 - **Win** - if player gets a row, horizontal, or diagonal
 - **Draw** - if neither player can achieve this
- Allows for a move if the cell is empty
- Checks all rows and columns in the board



LastMove Implementation

- Displays the last move made by the player
 - For example, if Player X makes a move here
 - Then LastMove displays that a move was made in row 1 column 1 to Player O

Player O turn		
X		

Last Move:
Player X made
a move to
row 1 col 1

TA Feedback

Deny a move if a player tries to make a move without a second person in the server:

- If there is only one person in the server, you cannot play the game
- Sets all the buttons to non-clickable
- Sets buttons back to normal when another person joins

